Small Business Profitability Analysis

Sales, profit, discounts and regional performance in retail transactions

Python | Pandas | Plotly | Interactive Dashboard

Project overview

1 Objective

Find the sales drivers and the loss-making areas of the business.

2 Dataset

Kaggle Superstore Final: 9,994 order lines, 2014–2017. A retail proxy, not a survey of small businesses.

3 Method

Clean data in pandas, compute KPIs, then animated Plotly charts and filter widgets.

4 Output

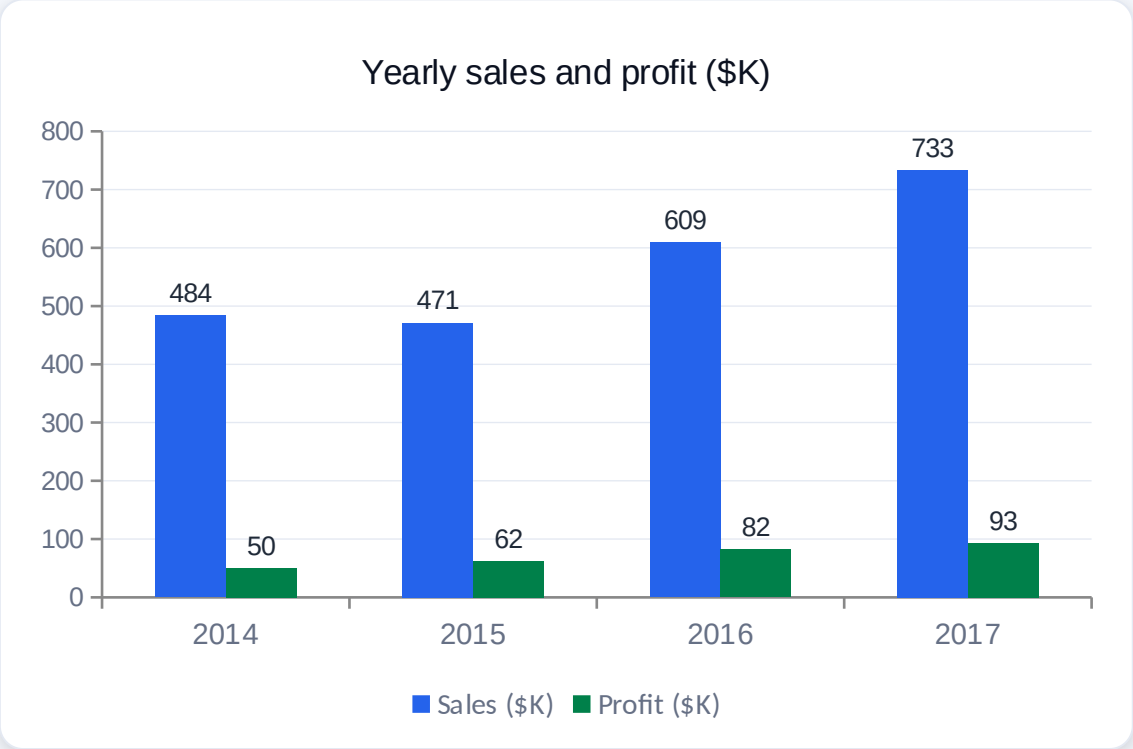
Interactive dashboard (category, region, segment filters) plus an HTML export.

Executive KPI overview

$2.30M	$286K	12.5%	5,009
Total sales	Total profit	Profit margin	Orders

1,871 order lines (18.7%) lost money. Healthy overall margin hides a loss-making tail that the rest of this deck tries to locate.

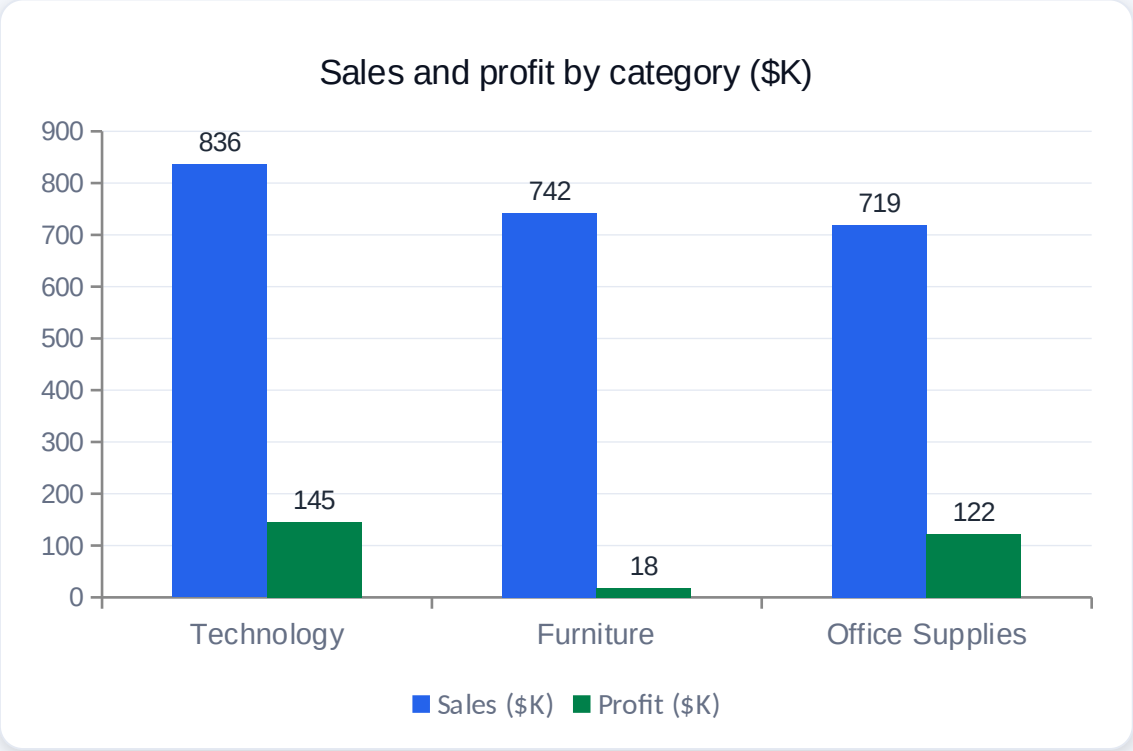
Sales and profit are growing



Sales up 51% from 2014 to 2017
Profit nearly doubled, from $50K to $93K.

The notebook's animated chart reveals this month by month; yearly totals shown here for clarity.

Furniture sells well but barely profits



Technology

17.4% margin: best earner

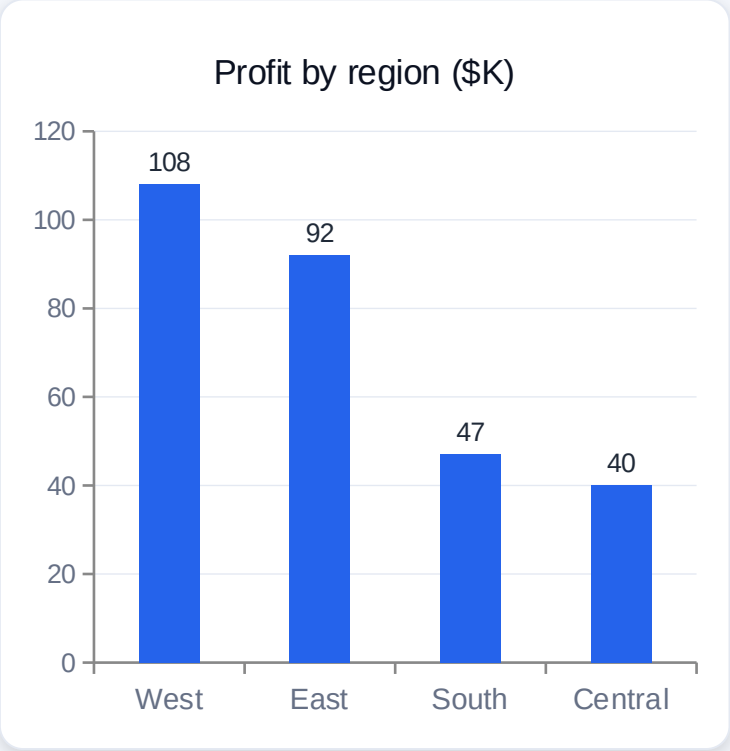
Office Supplies

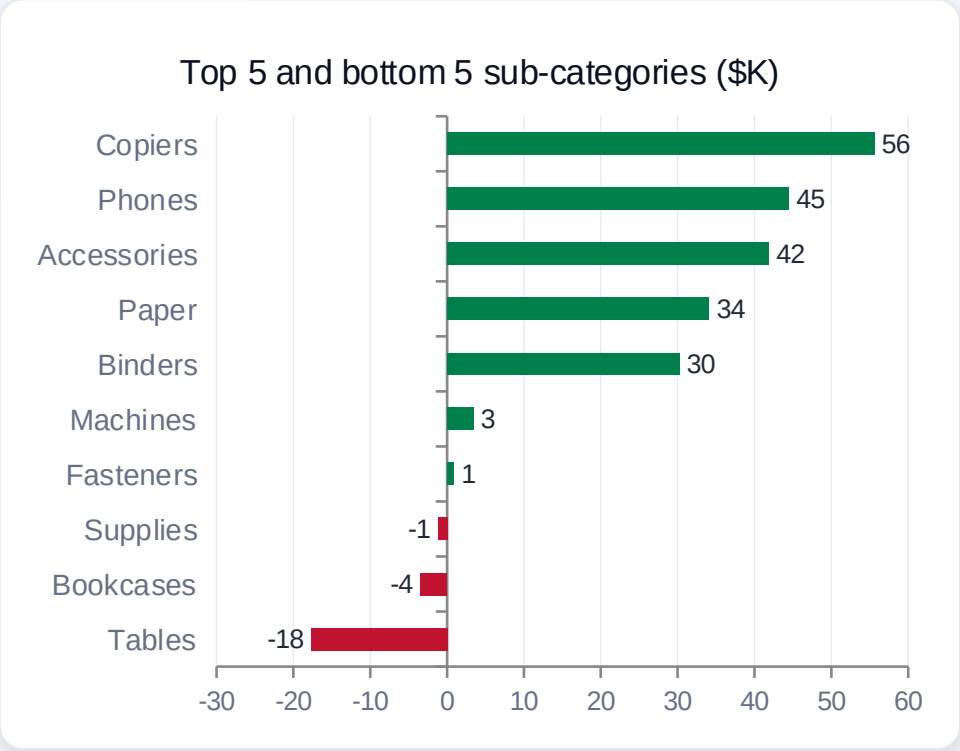
17.0% margin: steady

Furniture

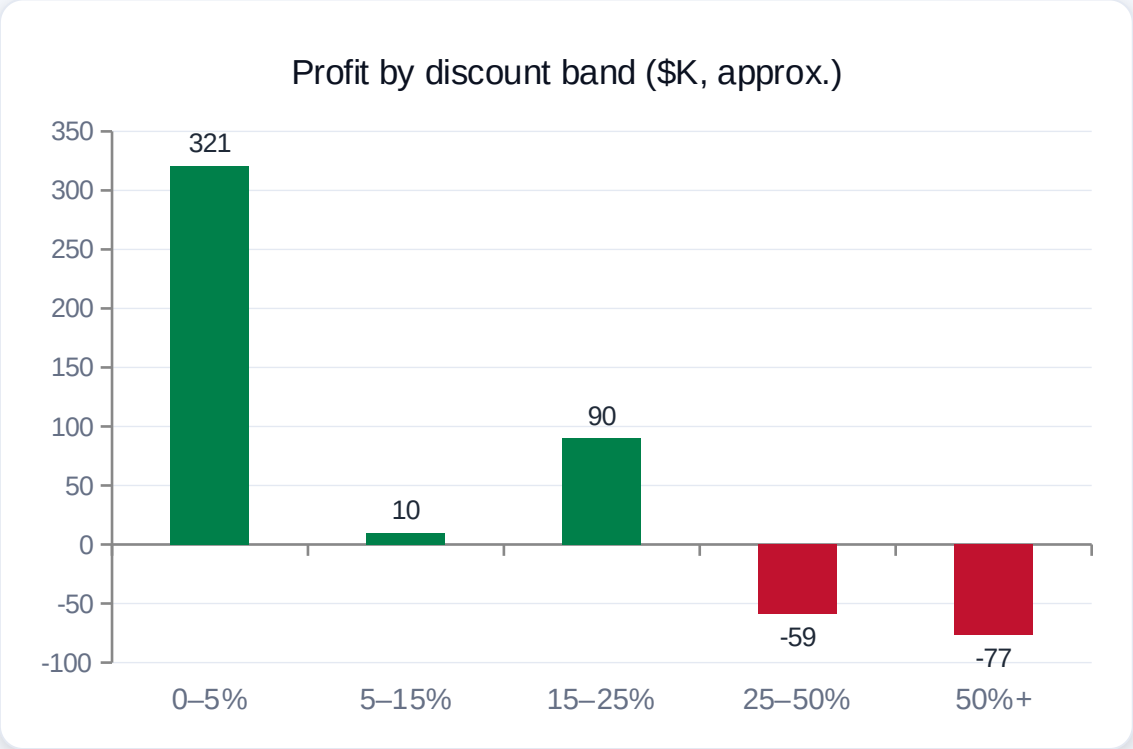
2.5% margin: weak

Where profit comes from, and where it leaks





Deep discounts coincide with losses



-0.22

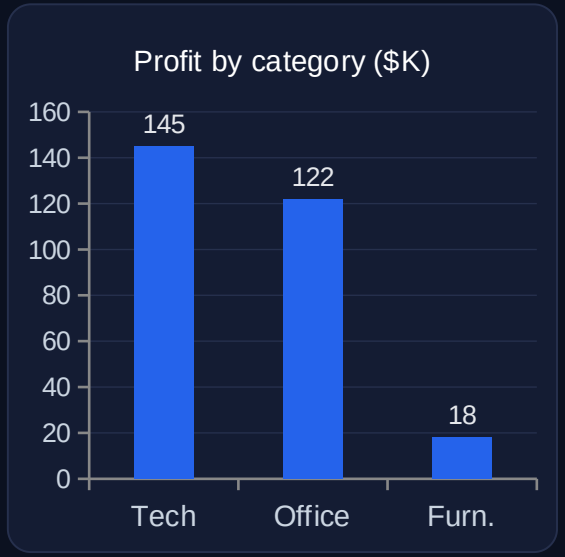
discount vs profit correlation

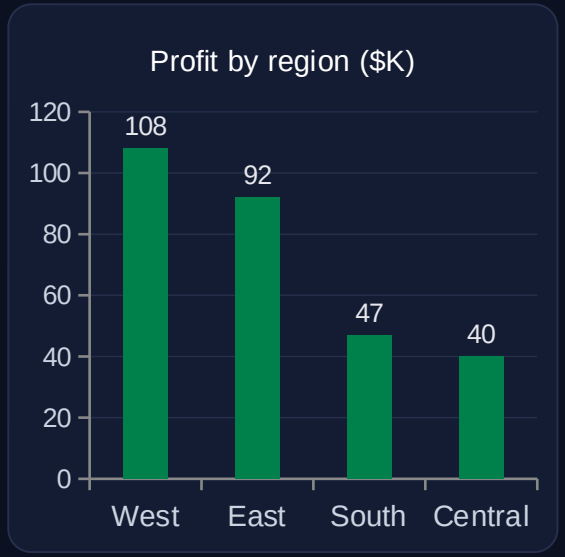
Discounts above 25% are net loss-making. This is an association, not proof that discounts alone cause the losses.

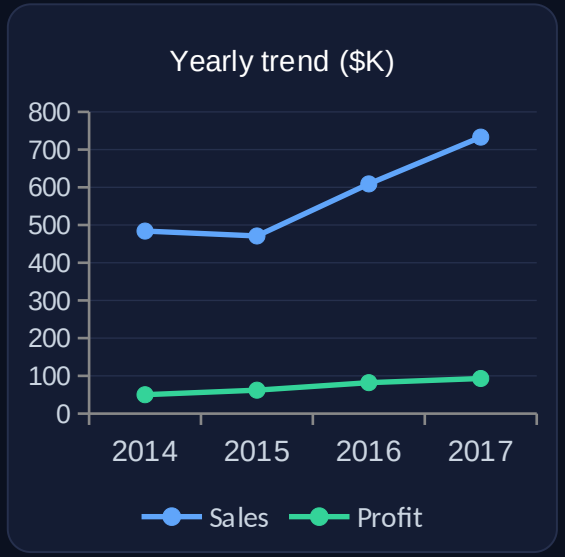
Interactive dashboard

Filters: Category | Region | Segment

SALES	PROFIT	MARGIN	LOSS ROWS
$2.30M	$286K	12.5%	1,871







Recommendations

1

Cap discounts

Test a 20–25% ceiling; deeper discounts are where losses concentrate.

2

Fix Tables and Bookcases

Review pricing, supplier cost and shipping for the two loss-making sub-categories.

3

Back the winners

Prioritise Technology, Copiers, Phones and the West and East regions.

Dataset is a retail proxy; findings describe transaction profitability, not small-business surveys.